

Georg von Westphalen

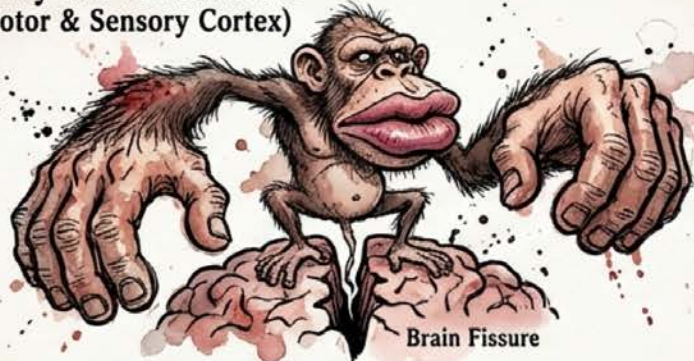
MedKayfab: Neuroanatomy Edition!

Uncle FrizzleBob's Krazy Kayfabizzaro Komik Kard Kult Klub!



Homy the Homunculus (Motor & Sensory Cortex)

STAT: MCA > ACA



Brain Fissure

POWER: MCA Stroke — Target's contralateral face and arm are paralyzed.
LORE: The leg representation descends into the interhemispheric fissure. Bilateral ACA territory ischemia mimics spinal cord disease.

Broca & Wernicke (The Language Lords)

STAT: 95% Left Hemisphere



POWER: Conduction Aphasia — Target speaks fluently but cannot repeat a single word.
LORE: Broca builds speech (expressive). Wernicke understands it (receptive).

Fronty the Frontal Lobe (Executive Function)

STAT: Area 8 (Frontal Eye Fields)



POWER: Frontal Release — Re-equips primitive grasp and rooting reflexes.
LORE: Irritative lesion (seizure) makes eyes look AWAY. Destructive lesion (stroke) makes eyes look TOWARD the lesion.

Parry the Parietal Lobe (Spatial Integration)

STAT: 2-Point Discrimination



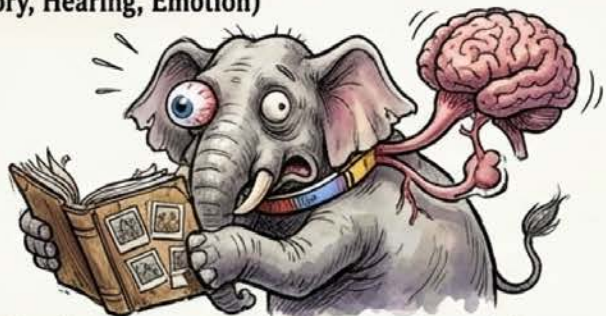
POWER: Hemispatial Neglect — The left half of the opponent's world ceases to exist.
LORE: Dominant side handles math and fingers (Gerstmann). Non-dominant side integrates the spatial map.

KayfaBIZARRO!

KayfaBIZARRO!!

Tempo the Temporal Lobe (Memory, Hearing, Emotion)

STAT: 72h (HSV-1 affinity)

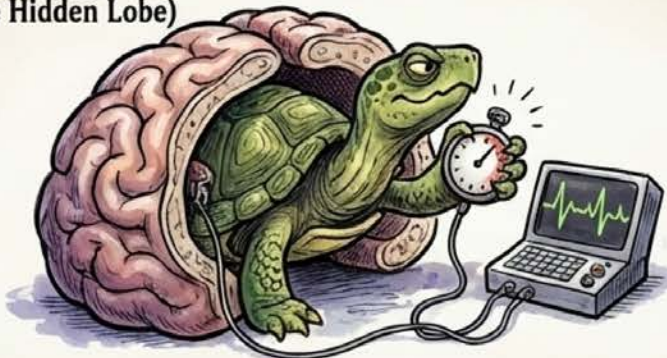


POWER: Uncal Herniation — Crush CN III to dilate the pupil, then cause false-localizing hemiplegia.

LORE: Hippocampus encodes memory. The Uncus herniates through the tentorium to cause the blown pupil.

Insula Izzy (The Hidden Lobe)

STAT: QT Prolongation

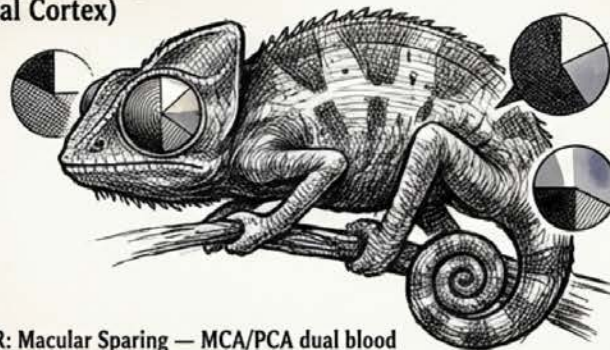


POWER: Autonomic Arrhythmia — Manipulate internal body state and cardiac rhythm.

LORE: Buried deep in the Sylvian fissure. Right insular stroke triggers sympathetic storms and arrhythmias.

Occi the Occipital Lobe (Visual Cortex)

STAT: Area 17 / V1



POWER: Macular Sparing — MCA/PCA dual blood supply protects central vision during PCA stroke.

LORE: Meyer's loop damage is pie in the sky (superior defect). Parietal optic radiation damage is pie on the floor (inferior).

Limby the Limbic System (Emotion & Memory)

STAT: Papez Circuit 1



POWER: Anterograde Amnesia — Opponent cannot form new declarative memories.

LORE: Mammillary body atrophy (Wernicke-Korsakoff) permanently breaks the Papez episodic memory circuit.

Kayfa BIZARRO!

HUMBUG!

Basso the Basal Ganglia STAT: D1 Excite / D2 Inhibit
(Movement Modulator)



POWER: Direct vs Indirect — Dopamine strengthens the direct (go) and weakens the indirect (stop).
LORE: Parkinson's loses dopamine (stuck in stop).
Huntington's loses the indirect pathway early (stuck in go).

Dorsal the Dorsal Column STAT: 100% Uncrossed
(Precision Sensory) in Spine



POWER: Romberg Fall — Opponent falls over the moment they close their eyes.
LORE: Fine touch and proprioception ascend ipsilaterally and do not cross until the medial lemniscus in the medulla.

Thalamo the Thalamus
(The Great Relay)

STAT: VPL / VPM



POWER: Déjerine-Roussy — Inflict burning, dysaesthetic pain weeks after a VPL infarct.
LORE: Every sensory modality except smell must synapse here before reaching the cortex.

Spino the Spinothalamic Tract STAT: 1-2 Segments
(Pain & Temp)



POWER: Cape Distribution — Syringomyelia expands the central canal, cutting the crossing fibers.
LORE: Pain and temperature cross the midline almost immediately in the anterior commissure of the spinal cord.

Cortico the Corticospinal Tract (Upper Motor Neuron)

STAT: 85–90%
Decussate



POWER: UMN Spasticity — Inflict hyperreflexia, spasticity, and a positive Babinski sign. No wasting.
LORE: The internal capsule is the bottleneck. A tiny lacunar infarct here causes massive, dense contralateral hemiplegia.

Cere the Cerebellar Pathways (Ipsilateral Coordination)

STAT:
2 Crossings = 0



POWER: Intention Tremor — Opponent's movements worsen as they approach the target.
LORE: Output crosses to the contralateral thalamus, then the corticospinal tract crosses back. Net result: ipsilateral control.

Russel the Rubrospinal (Brainstem Motor)

STAT:
Red Nucleus



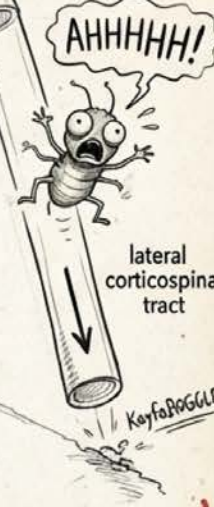
POWER: Decerebrate vs Decorticate — Dictate the opponent's coma posture based on lesion height.
LORE: Cortical loss = decorticate (arms flexed). Lesion below red nucleus = decerebrate (all limbs extended).

Spino the Spinal Cord Cross-Section

STAT:
ASA = Anterior 2/3



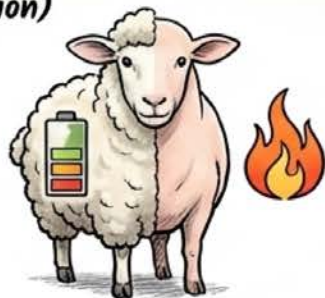
POWER: ASA Syndrome — Paraplegia and pain loss below lesion, but proprioception survives.
LORE: The anterior spinal artery feeds motor and pain tracts. The posterior spinal arteries feed the dorsal columns.



Keyhole Syndrome

Brown the Brown-Séquard (Hemisection)

STAT: 50% Cut



POWER: The Cleanest Cut — Ipsilateral motor/proprioception loss, contralateral pain loss.

LORE: The ultimate test of crossing rules. Pain crosses early, motor and touch cross late.

Middy the Midbrain

STAT: CN III & IV



POWER: Weber's Strike — Ipsilateral CN III palsy + contralateral hemiplegia.

LORE: Houses the cerebral peduncles and oculomotor nerves. Parinaud's syndrome (pineal tumor) blocks upgaze.

Conus the Conus & Cauda (Cord Termination)

STAT: S3-S5



POWER: Sphincter Sabotage — Saddle anesthesia and loss of anal tone demands urgent surgery.

LORE: Conus = UMN signs mixed with LMN bladder. Cauda Equina = Pure LMN flaccid weakness and areflexia.

Ponty the Pons

STAT: MLF & PPRF



POWER: Locked-In Syndrome — Bilateral ventral destruction leaves only vertical gaze and blinking.

LORE: The center for horizontal gaze. An MLF lesion causes Internuclear Ophthalmoplegia (INO).

Meddy the Medulla (The Vital Core)

STAT: PICA vs ASA



POWER: Wallenberg's Curse (Lateral Medulla) — Ipsilateral face pain, Horner's, hoarseness, and ataxia.
LORE: Controls breathing (pre-Bötzinger) and swallowing (nucleus ambiguus). Lesions here are life-threatening.

Cranio I-IV (The Anterior Four)

STAT:
CN II IN / III OUT



POWER: Pupillary Light Reflex — CN II senses the light, bilateral Edinger-Westphal nuclei send CN III to constrict.
LORE: A painful third nerve palsy with a blown pupil is a posterior communicating artery aneurysm emergency.

Cranio V-VII (The Face Specialists)

STAT:
72h Prednisolone



POWER: Bell's vs Stroke — LMN VII palsy drops the whole face: UMN spares the forehead.
LORE: CN VI has the longest intracranial course. An isolated CN VI palsy is a false localizing sign for raised ICP.

Cranio VIII-XII (The Lower Family)

STAT:
CP Angle

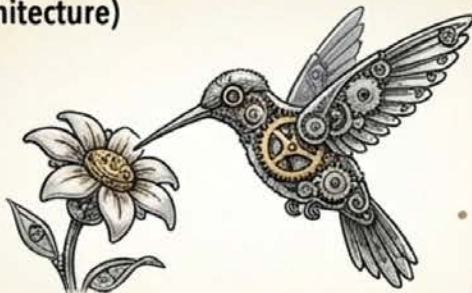


POWER: Jugular Foramen Mass — Compress IX, X, and XI to destroy swallowing and shoulder shrug.
LORE: Uvula deviates AWAY from a CN X lesion. Tongue deviates TOWARD a CN XII LMN lesion.

HUMBUG!

The Cerebellum (Architecture)

STAT: FDGE (Deep Nuclei)



POWER: Archicerebellum Spin — Flocculonodular lobe lesions destroy balance and trigger nystagmus.
LORE: Vermis controls the trunk (paleocerebellum). Lateral hemispheres control distal limbs (neocerebellum).

Brachy the Brachial Plexus (Roots to Branches)

STAT: C5-T1



POWER: Waiter's Tip (Erb's Palsy) — C5/C6 upper trunk avulsion during shoulder dystocia.
LORE: Lower trunk (C8/T1) injury causes Klumpke's claw hand PLUS Horner syndrome from T1 sympathetic drag.

Vertigo Vera (Central vs Peripheral)

STAT: HINTS > MRI in 24h



POWER: Test of Skew — Differentiate a benign inner ear from a posterior circulation stroke.
LORE: Pure vertical nystagmus is ALWAYS central. Direction-changing horizontal nystagmus is central.

Radi the Radial Nerve (The Extensor)

STAT: Saturday Night Palsy



POWER: Spiral Groove Crush — Target loses wrist and finger extension, but preserves elbow extension.
LORE: The triceps motor branch is given off BEFORE the nerve wraps around the humeral shaft.

KoyfaBIZARRO!

Ulna the Ulnar Nerve STAT: Medial 1.5 Fingers
(Intrinsic Hand)



POWER: Ulnar Paradox — A wrist lesion creates a WORSE claw than an elbow lesion.
LORE: A high lesion paralyzes FDP, meaning the fingers can't flex to form the full claw.

Medi the Median Nerve STAT: LOAF Muscles
(Precision Hand)



POWER: Carpal Tunnel Crush — Nocturnal paraesthesia in the lateral 3.5 digits.
LORE: Thenar wasting means the nerve is dying. Benediction hand only happens with a high (elbow) lesion.

Femmo the Femoral Nerve STAT: L2-L4
(The Quadriceps)



POWER: Psoas Hematoma — Compress the nerve within the psoas to kill hip flexion and knee extension.
LORE: Absent knee jerk + positive femoral stretch test + anterior thigh numbness.

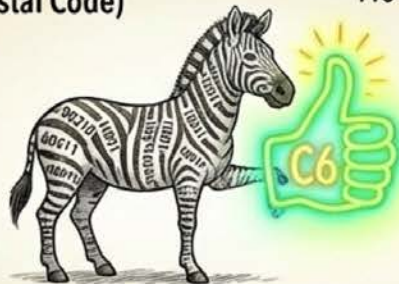
Sciati the Sciatic/Peroneal STAT: Fibular Neck
(The Leg)



POWER: Foot Drop — Common peroneal palsy loses dorsiflexion and eversion.
LORE: How to tell peroneal palsy from L5 root lesion? L5 also loses inversion (tibialis posterior).

Derma the Dermatomes (The Postal Code) STAT: T4=Nipple / T10=Umbilicus

STAT: T4=Nipple /
T10=Umbilicus



POWER: Sensory Level — Pinpoint the exact spinal segment of a cord transection.
LORE: The thumb is C6. The middle finger is C7. The pinky is C8. Always.

Reflex Rex

(Deep Tendon Reflexes)

STAT: 1a Afferent



POWER: Jendrassik Manoeuvre — Pull clasped hands apart to amplify gamma motor neuron activity.
LORE: Hyperreflexia + Clonus + Babinski = UMN lesion.
 Areflexia + Wasting = LMN lesion.

Autonomo the ANS STAT: T1-L2 vs Craniosacral
(Sympathetic vs Parasympathetic)

STAT: T1-L2 vs Craniosacral



POWER: Adrenal Bypass — The adrenal medulla acts as a modified sympathetic ganglion, releasing directly to blood.
LORE: Sympathetic preganglionics are short.
 Parasympathetic preganglionics are long.

Horner the Horner (Sympathetic Chain)

STAT: 3 Neurons



POWER: Hydroxyamphetamine Test — Dilates central and preganglionic lesions, fails in postganglionic.
LORE: Ptosis, Miosis, Anhidrosis. An acute Horner with neck pain is a carotid dissection until proven otherwise.

Derma Root Pain (Radiculopathy)

STAT: L4/L5 Disc → L5 Root



POWER: Spurling's Shock – Axial compression of rotated neck reproduces cervical radicular pain.
LORE: Posterolateral disc herniations in the lumbar spine compress the exiting nerve root ONE LEVEL BELOW the disc.

Peripheral Percy (Neuropathy Patterns)

STAT: Albumino-cytological



POWER: Dying-Back (Length-Dependent) – The longest axons fail first, causing stocking-glove sensory loss.
LORE: Guillain-Barre Syndrome shows high CSF protein with normal cell counts (dissociation) and absent reflexes.

Ventri the Ventricular System (CSF Circulation)

STAT: 500 mL / Day



POWER: Hakim's Triad – Normal Pressure Hydrocephalus causes wobbly, wacky, and wet (gait, dementia, incontinence).
LORE: Aqueductal stenosis is obstructive. Post-SAH failure of arachnoid granulations is communicating.

Meningo the Meninges (The Wrapper)

STAT: 3 Layers



POWER: Lucid Interval – Epidural hematoma patients wake up, talk, and then suddenly herniate.
LORE: Epidural = Arterial (Middle Meningeal), Biconvex. Subdural = Venous (Bridging Veins), Crescent.

Foramen of Monro

Foramen of Monro

Hernia the Herniation

STAT: Kernohan's Notch

(Syndromes)



POWER: False Localization — Contralateral peduncle pushed against tentorial notch causes **IPSILATERAL** hemiplegia.
LORE: A fixed dilated pupil is the earliest reliable clinical sign of uncal herniation (parasympathetic fibers on outside of CN III).

Blood Brain Barry

(The Barrier)

STAT: GLUT1 Transport



POWER: Vasogenic Edema — Bacterial meningitis breaks the tight junctions; dexamethasone seals them.
LORE: Tumor vasculature lacks a blood-brain barrier, which is why gadolinium contrast lights up tumors on MRI.

Cerebral Vera Vasculature

STAT: Lenticulostriate

(The Supply)



POWER: Subclavian Steal — Arm exercise reverses vertebral artery flow, causing posterior circulation ischemia.
LORE: Sudden unilateral deafness + facial numbness + contralateral body numbness = AICA stroke.

Aneurysm Annie

(The Ticking Bomb)

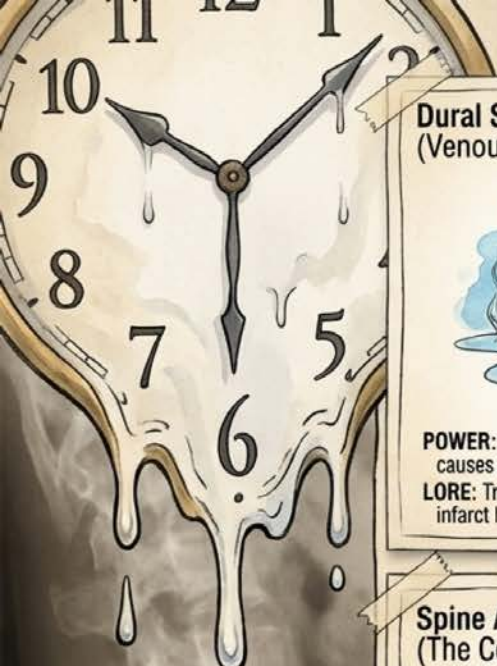
STAT: 85% Saccular



POWER: Thunderclap Headache — Saccular aneurysm rupture into subarachnoid space.
LORE: A painful CN III palsy is a Posterior Communicating Artery (PCoA) aneurysm until proven otherwise.

What the FLUFF?!





Dural Sinus Derek (Venous Drainage)

STAT: Empty Delta Sign



POWER: The Great Mimic — Cerebral Venous Thrombosis causes raised ICP, seizures, and focal deficits.

LORE: Treat with anticoagulation (LMWH) even if the venous infarct has already caused hemorrhagic transformation.

Spino Cord Syndromes Summary

STAT: Central Cord > Arms

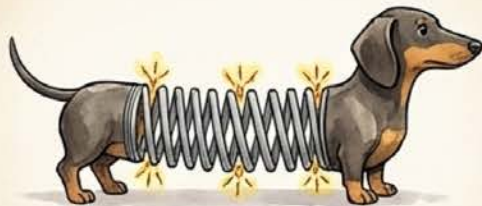


POWER: Central Cord Syndrome — Hyperextension in elderly patients destroys the central cervical cord.

LORE: Arm motor fibers lie centrally, leg fibers lie peripherally. Therefore, arms are much weaker than legs.

Spine Anatomy Sandy (The Column)

STAT: C6 exits C5/C6



POWER: Neurogenic Claudication — Spinal stenosis causes bilateral leg pain when walking, relieved by bending forward.

LORE: In the cervical spine, the root exits ABOVE its numbered vertebra. In the lumbar spine, it exits BELOW.

Sleep Anatomy Sam (The Flip-Flop Switch)

STAT: HLA-DQB1*06:02



POWER: REM Intrusion — Loss of orexin drops the patient directly from wakefulness into atonic REM sleep (Cataplexy).

LORE: The VLPO and ARAS mutually inhibit each other. Orexin from the lateral hypothalamus stabilizes the awake state.

HUMBUG!



HUMBUG!



Pain Percy (Nociception & Modulation)

STAT: A-delta / C fibers



POWER: Gate Control Theory — Activate large A-beta touch fibers to inhibit C fiber pain transmission.

LORE: Referred pain happens because visceral and somatic afferents converge on the exact same dorsal horn neurons.

Visual Percy (Optic Radiations)

STAT: Meyer's Loop



POWER: Pie in the Sky — Anterior temporal lobectomy damages Meyer's loop, causing superior quadrantanopia.

LORE: The macula occupies 50% of the primary visual cortex, explaining central vision sparing in PCA strokes.

Eye Move Eddie (Conjugate Gaze)

STAT: Area 8 → PPRF



POWER: One-and-a-Half Syndrome — PPRF + MLF lesion on same side. Only contralateral abduction survives.

LORE: MLF lesions cause INO (adduction failure). Bilateral INO in a young patient is Multiple Sclerosis until proven otherwise.

Memory Mike (Declarative Systems)

STAT: Papez Circuit



POWER: Transient Global Amnesia — Sudden anterograde amnesia caused by venous backflow hitting the hippocampus.

LORE: Hippocampus is essential for encoding. Remote memories survive hippocampal destruction due to systems consolidation.

Hypothala the Hypothalamus (Homeostasis)

STAT: SLAM DR Nuclei



POWER: Pan-hypopituitarism — Craniopharyngioma crushes the stalk, destroying endocrine and autonomic control.

LORE: Lateral hypothalamus = hunger (orexin). Ventromedial = satiety. Suprachiasmatic = master clock.

Pituitary Pete (The Sella Turcica)

STAT: >10mm = Macroadenoma



POWER: Apoplexy — Sudden hemorrhage of adenoma causes thunderclap headache and visual loss.

LORE: Suprasellar extension hits crossing nasal fibers, causing bitemporal hemianopia (tunnel vision).

The Neuroanatomy Attending (Master Review)

STAT: 3 Patients,
0 Approximations



POWER: Localize. Name. Explain. — Map the Horner's, the ataxia, and the hoarseness to the exact vessel.

LORE: You don't know it until you can trace the wiring backward from the deficit to the lesion without looking it up.

Uncle FrizzleBob's Rule of the Street

STAT: 0% Filler



POWER: Epistemic Sabotage — Destroy the boring format to perfectly preserve the factual integrity.

LORE: You remember stories, not lists. The one who knows where a structure lives can predict what happens when it dies.

STAY FLUFFY!